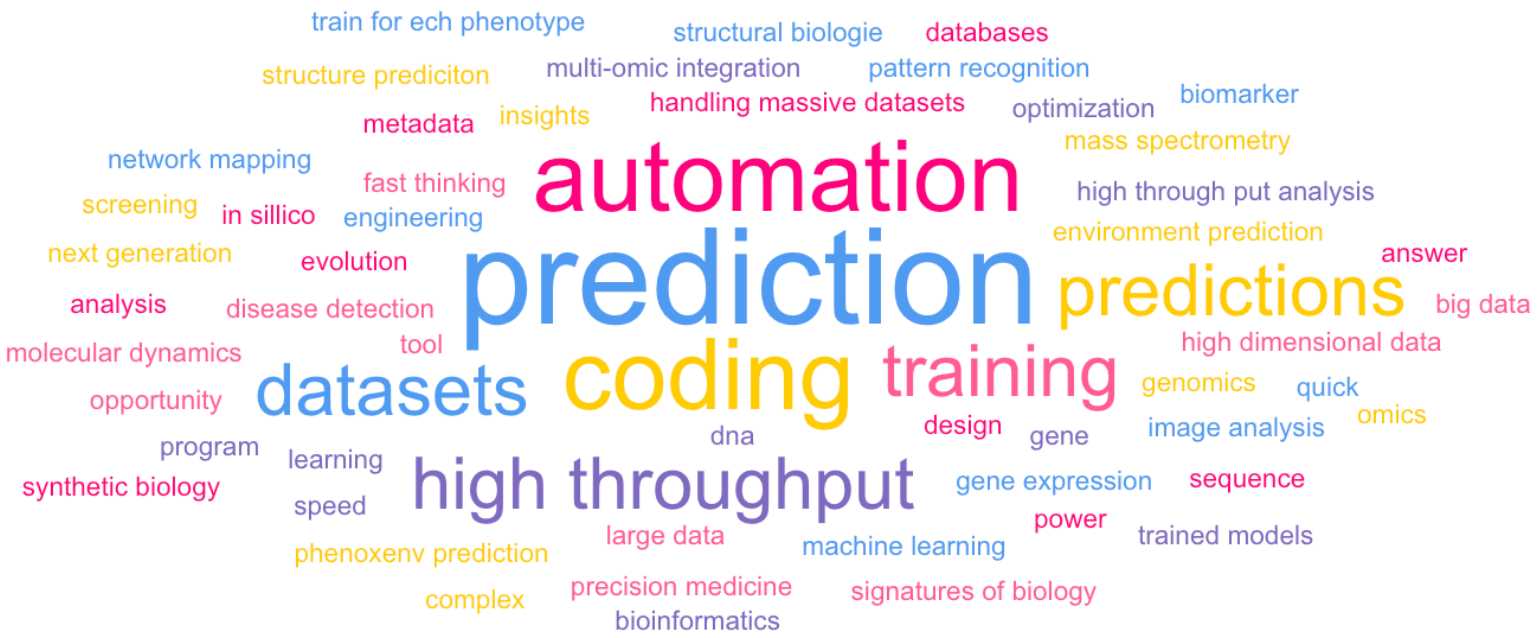
Introduction to Machine Learning

How to Think about Biology as a Machine Learning Problem?

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University of Bath
15/06/2026

What comes to mind when you hear “AI for Biology”?



AI for Science



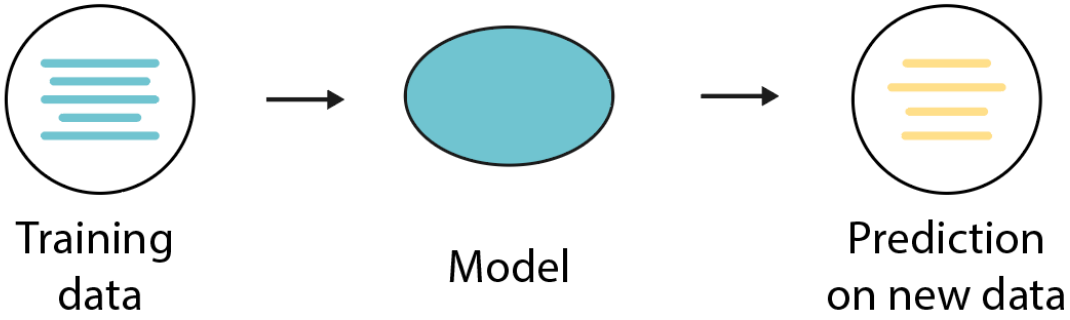
Credit: https://medium.com/@kenji-onisuka/ai-takes-the-nobel-prize-its-impact-on-science-and-society-4c73437fbbf8

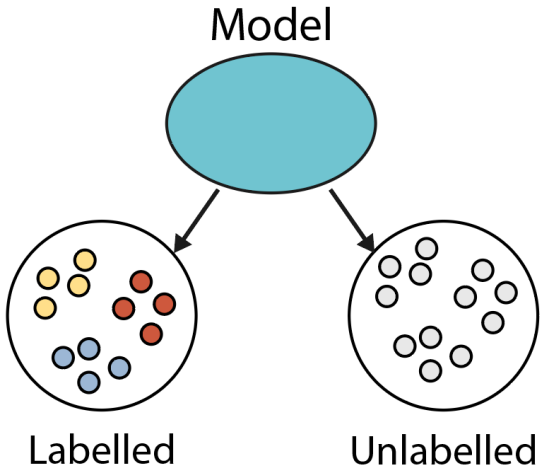
Outline

- Why biology needs AI
- Turning biological questions into machine learning problems
- The challenges of biological data
- Learning from data: unsupervised and supervised approaches
- Case studies: cells, drugs, and images
- Lessons learned: what ML can (and can’t) do in biology

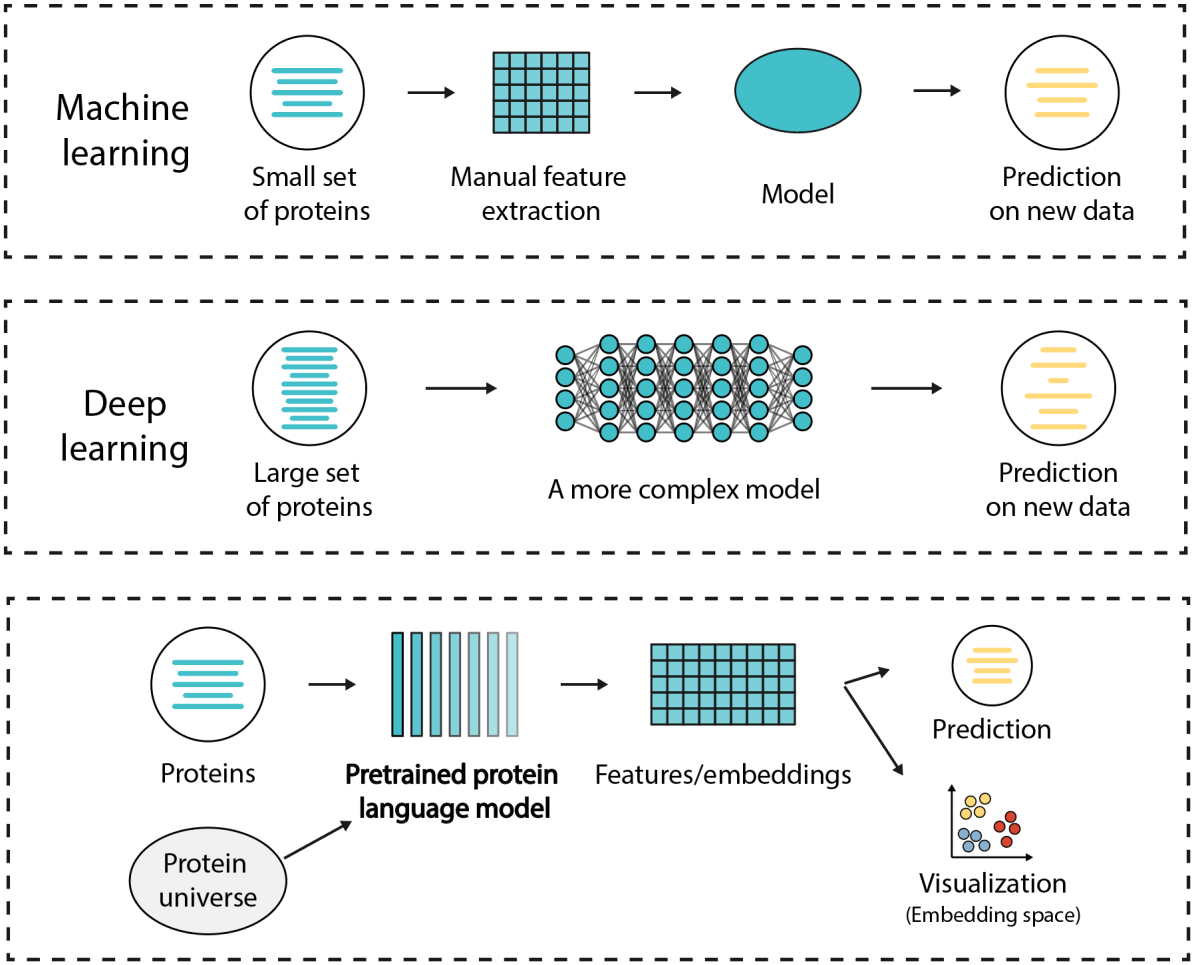
What Is Machine Learning?

- Learn patterns from data
- Input: numeric format (vectors, matrices)
- Output:
  - Supervised: prediction with labels
  - Unsupervised: find structure in unlabeled data

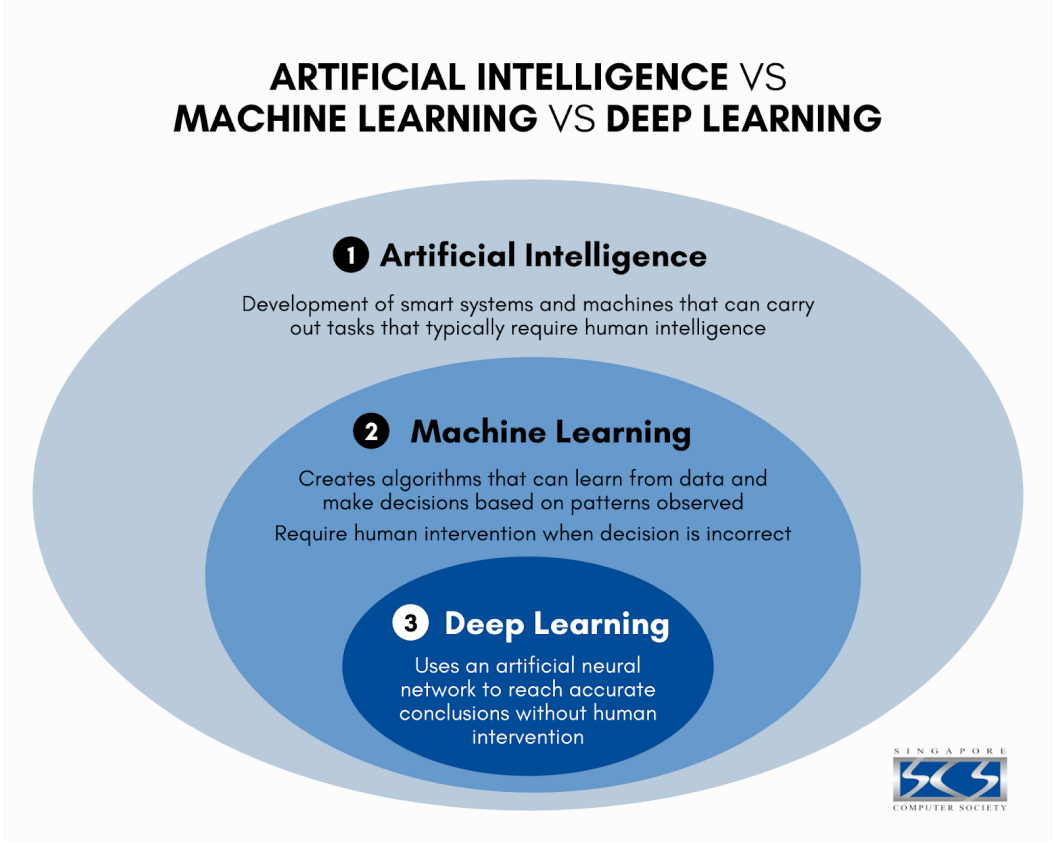




Key Machine Learning Concepts



AI vs Machine Learning vs Deep Learning

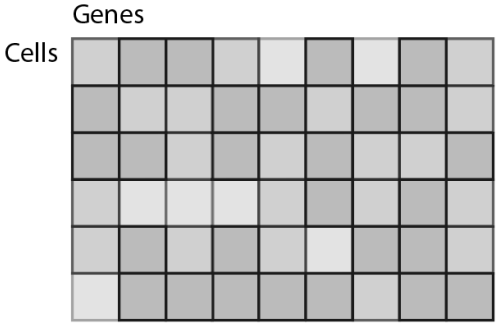


Credit: https://www.scs.org.sg/articles/machine-learning-vs-deep-learning

Why Biology Needs Machine Learning

- Complex data: omics, imaging, time-series
- Pattern recognition beyond human intuition
- ML helps classify, predict, discover

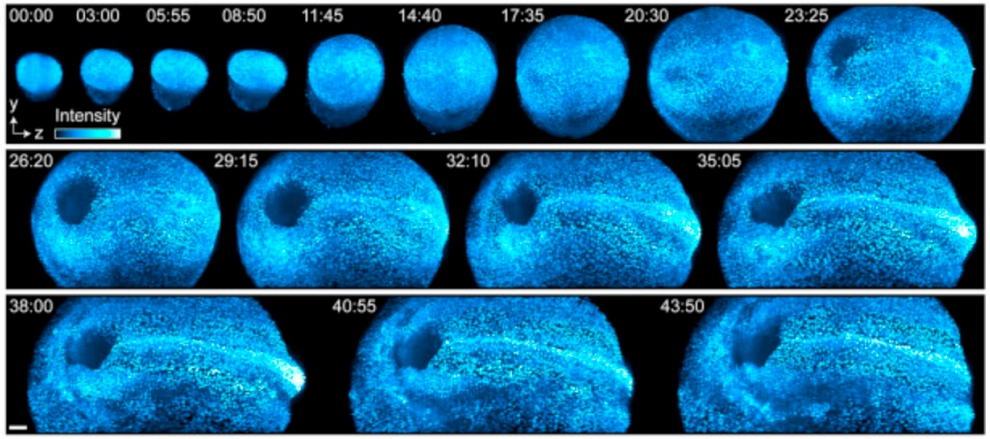
Gene expression matrix:



Protein sequences:

MKKLLPILIGLSLSGFSSLSQAENLMQVYQ
QARLSNPELRKSAADRDAAFEKINEARSP
LLPQLGLGADYTYSNGYRDAN…

Developing embryos:



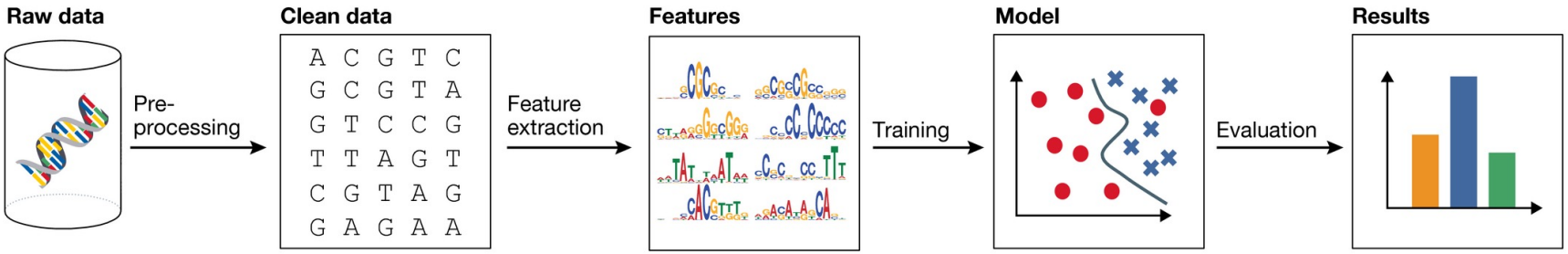
McDole et al., Cell (2018)

The Core Question

- **"Can we turn a biological problem into a machine learning problem?"**
- Examples:
  - Predict cell types from gene expression
  - Forecast drug responses
  - Image processing

The ML Pipeline

- 1. Data collection
- 2. Preprocessing & normalisation
- 3. Feature engineering
- 4. Model selection
- 5. Training
- 6. Evaluation
- 7. Interpretation & biological insight



Credit: https://www.embopress.org/doi/full/10.15252/msb.20156651

Biological Data Are Hard

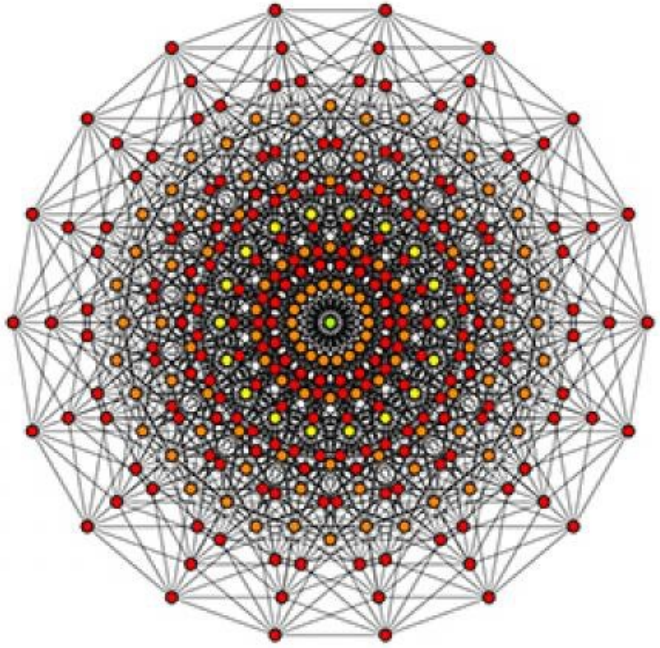
- High dimensional, low sample size ($p \gg n$)
- Batch effects
- Noise & dropout (scRNA-seq)
- Non-linear relationships
- Imbalanced classes



Credit: https://fortune.com/2018/03/19/big-data-digital-health-tech/

High-Dimensionality Problem

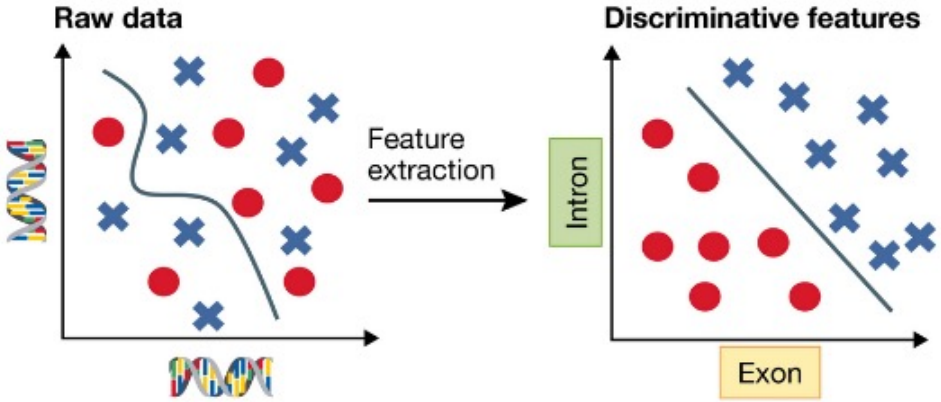
- Thousands of genes, few samples
- Hard to visualize or model directly
- Curse of dimensionality
- Leads to overfitting without dimensionality reduction



Credit: https://www.cam.ac.uk/research/news/lost-in-high-dimensional-space-study-improves-the-cure-for-the-curse-of-dimensionality

Feature Preprocessing

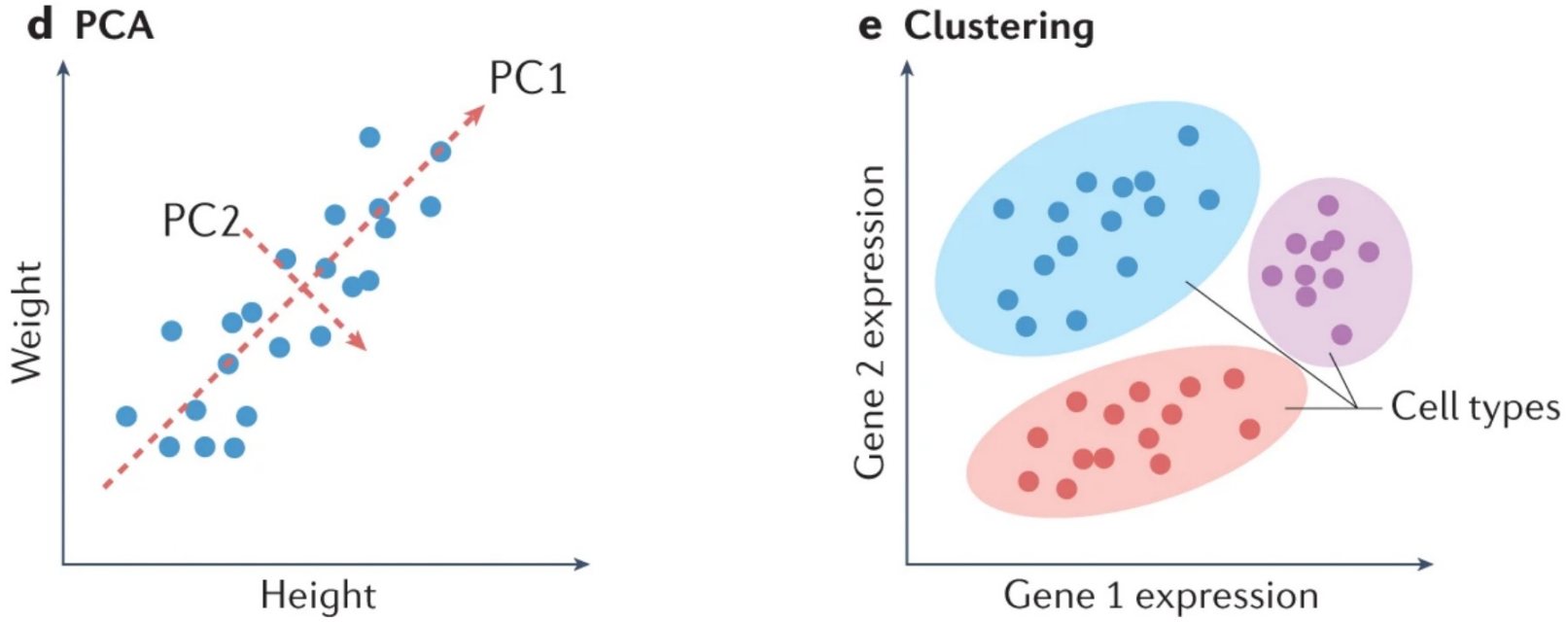
- Normalisation (log, scaling)
- Standardisation (z-score)
- Highly variable gene selection
- Handling missing values
- Ensures ML models behave properly



Credit: https://www.embopress.org/doi/full/10.15252/msb.20156651

Unsupervised Learning

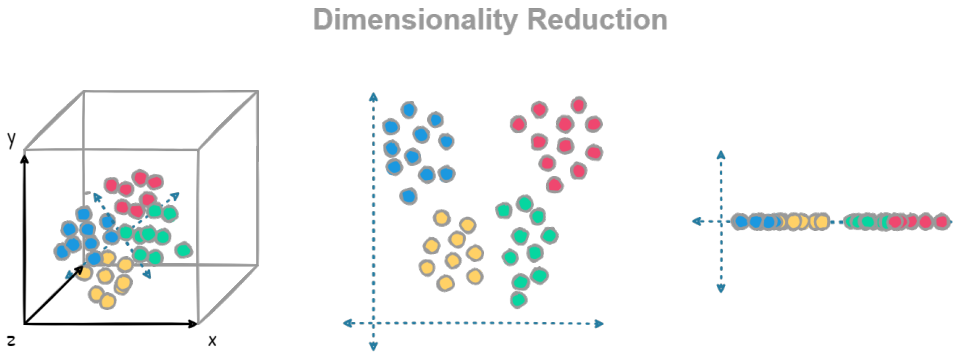
- Finding structure without labels
  - Dimensionality reduction (e.g., PCA, t-SNE and UMAP)
  - Clustering (e.g., k-Means and Hierarchical Clustering)



Credit: https://www.nature.com/articles/s41580-021-00407-0

What Is Dimensionality Reduction?

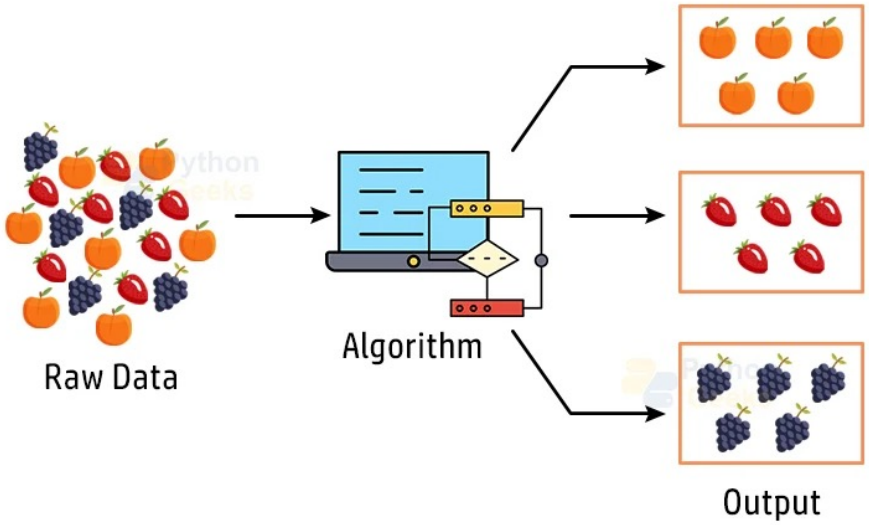
- Goal: project high-dimensional data into a lower-dimensional representation while preserving meaningful structure.
- Used for:
  - Visualization
  - Noise reduction
  - Preprocessing before clustering
  - Feature extraction for ML models



Credit: https://blog.roboflow.com/what-is-dimensionality-reduction/

Clustering Algorithms

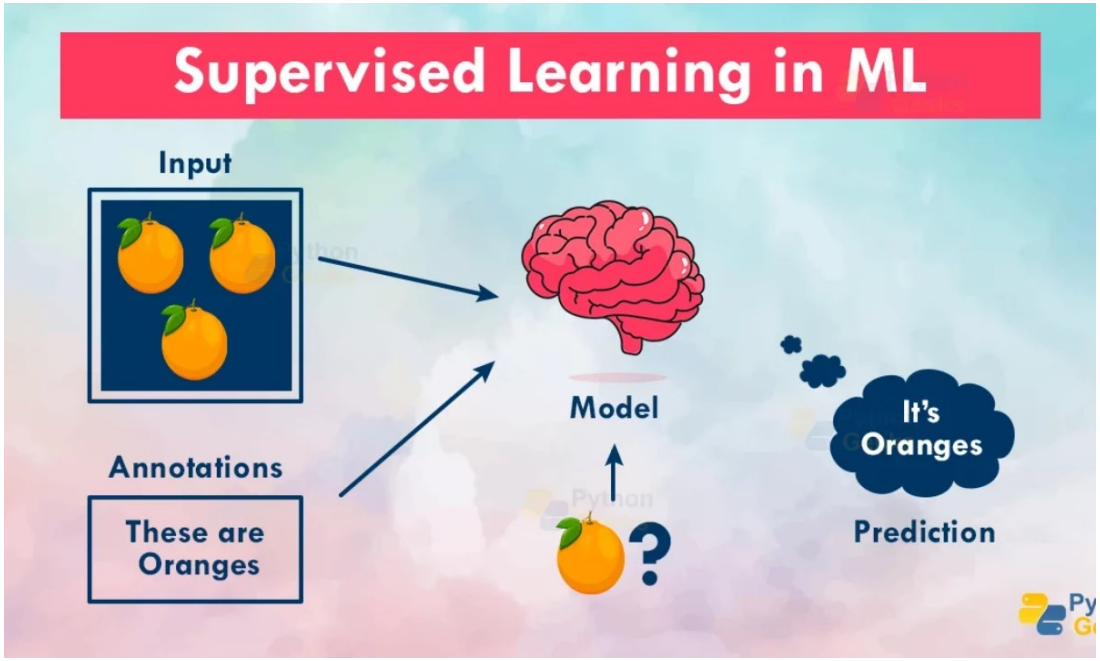
- Two main types:
  - Partitioning methods (k-means)
  - Hierarchical methods (HCA)
- Key difficulty:
  - No “ground truth” labels → evaluation is harder.



Credit: https://pythongeeks.org/clustering-in-machine-learning/

Supervised Learning

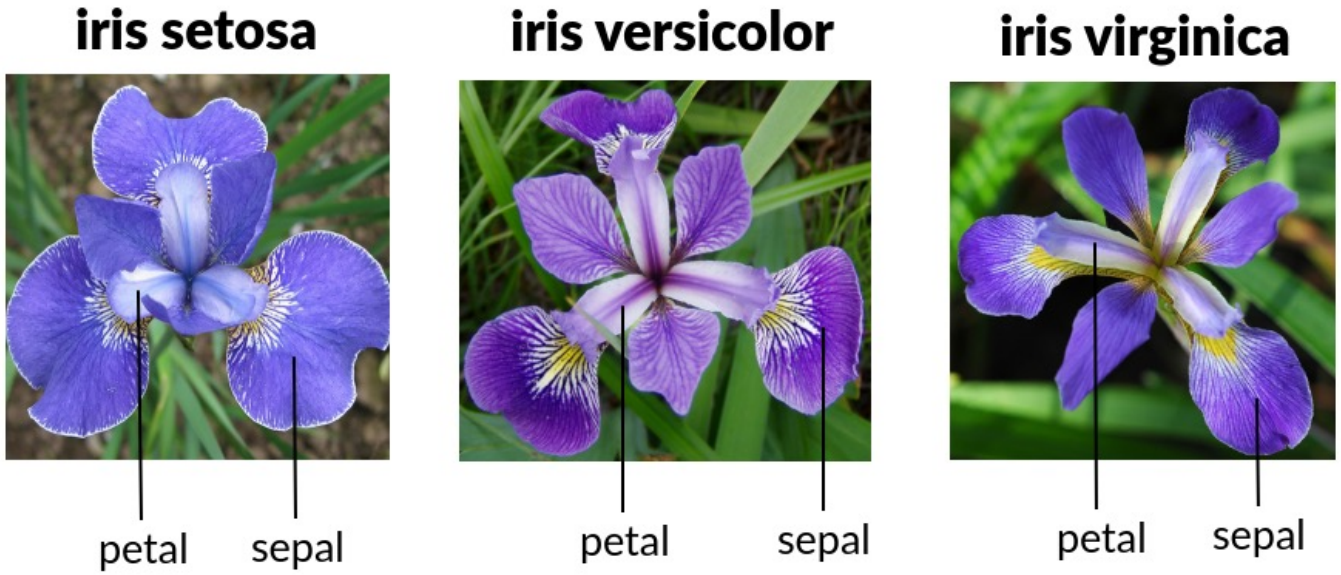
- Goal: learning from labelled data.
  - Predicting classes
  - Predicting continuous outcomes
  - Ranking & scoring tasks



Credit: https://pythongeeks.org/supervised-learning-in-machine-learning/

Classification Examples

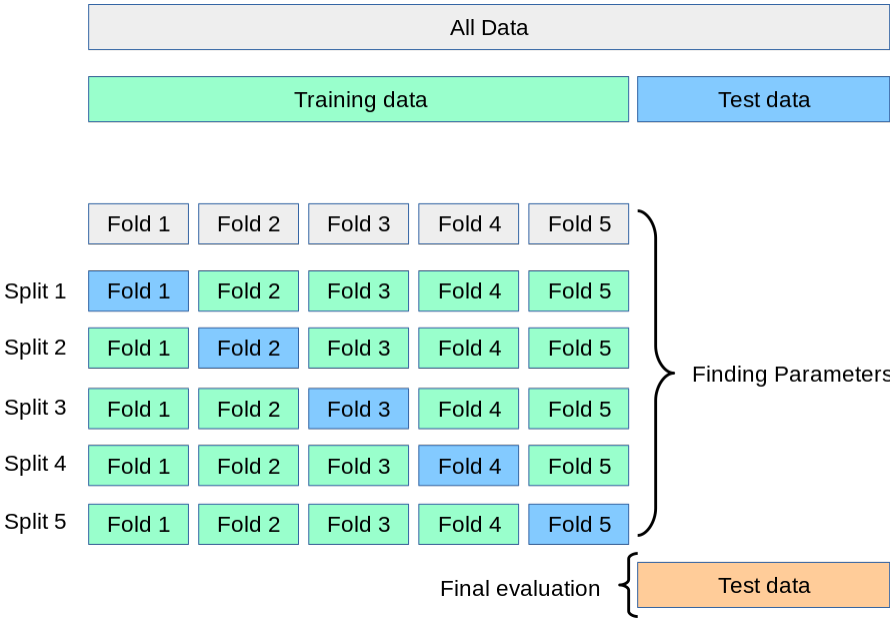
- Iris dataset:
  - Inputs: sepal length, sepal width, petal length, and petal width.
  - Output: species



Credit: https://www.analyticsvidhya.com/blog/2022/06/iris-flowers-classification-using-machine-learning/

Train–Test Split

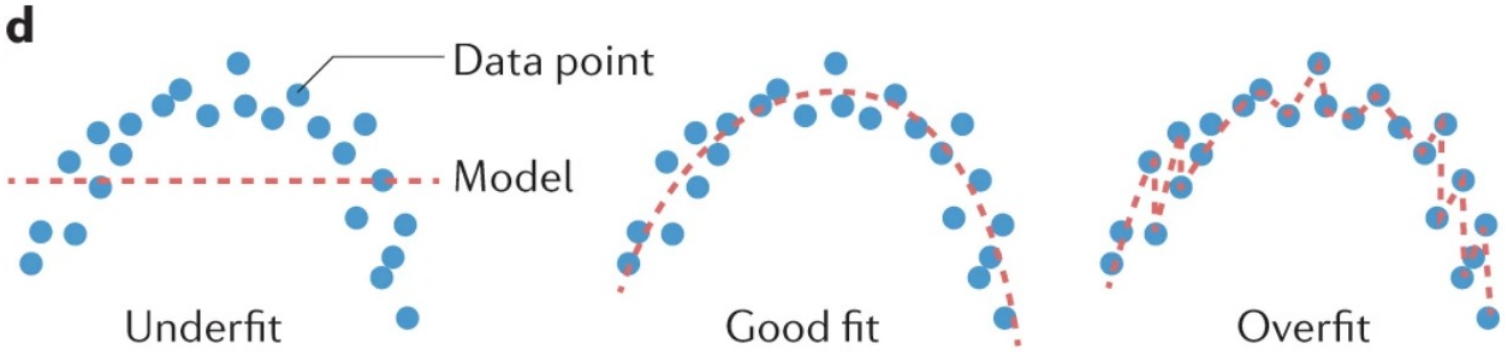
- To avoid overfitting:
  - Training set
  - Test set
  - Sometimes validation set
  - Cross-validation for small datasets



Credit: https://scikit-learn.org/stable/modules/cross_validation.html

Overfitting vs Underfitting

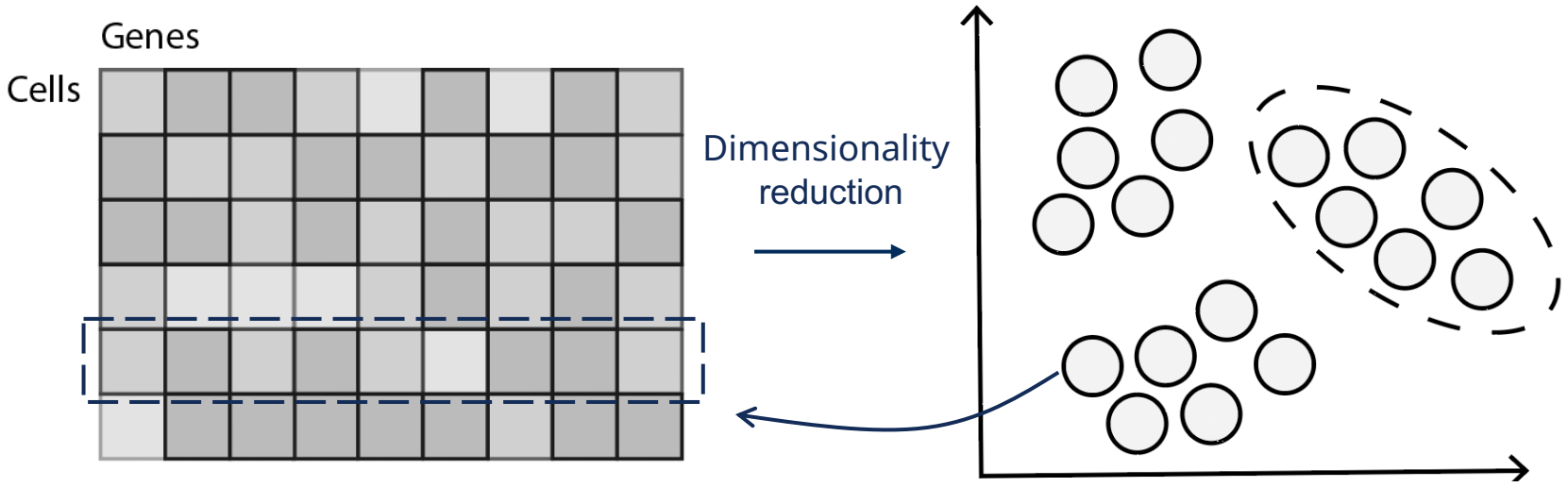
- Overfitting: model learns too much
- Underfitting: model learns not enough
- Goal = balance complexity vs generalization



Credit: https://www.nature.com/articles/s41580-021-00407-0

Case Study #1 — Single-Cell RNA-seq

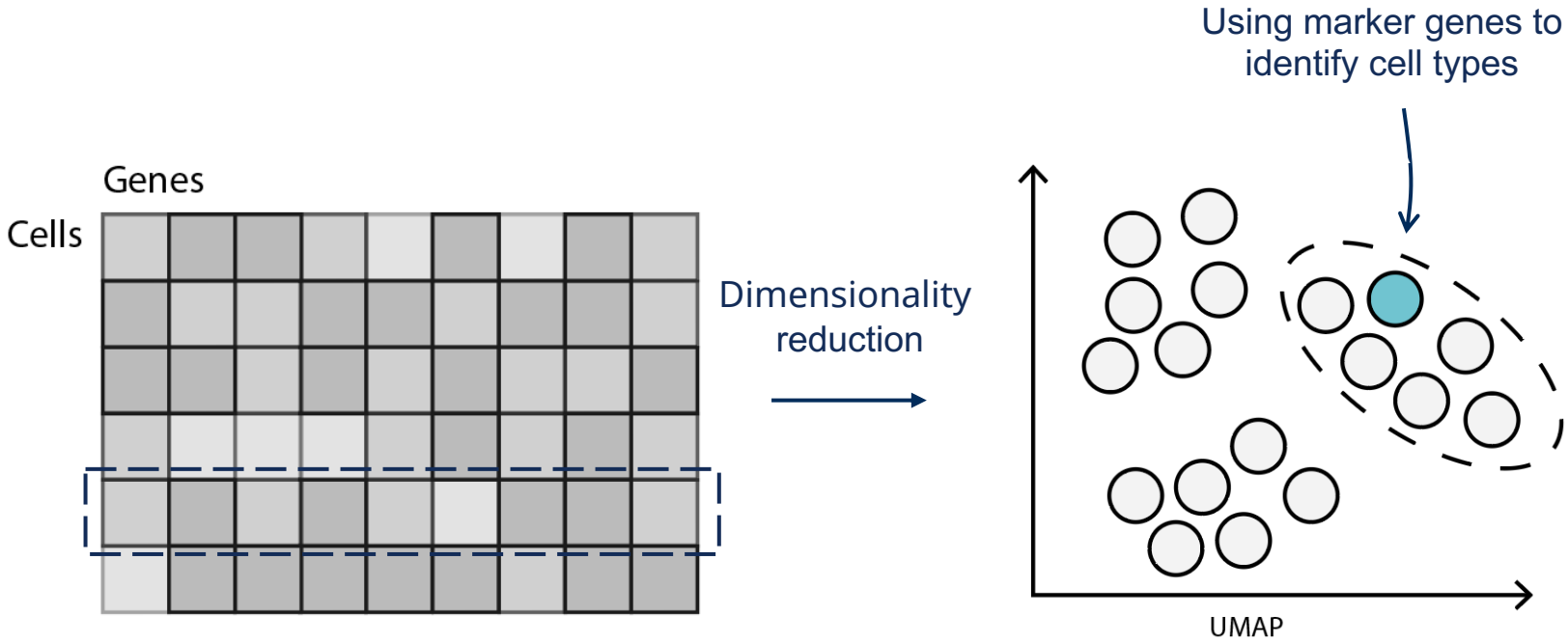
- Goal: Identify cell types
- Data: Expression vectors without cell type labels
- Task: Clustering (unsupervised)



Uniform Manifold Approximation and Projection (UMAP): a machine learning algorithm used for dimensionality reduction.

Case Study #1 — Single-Cell RNA-seq

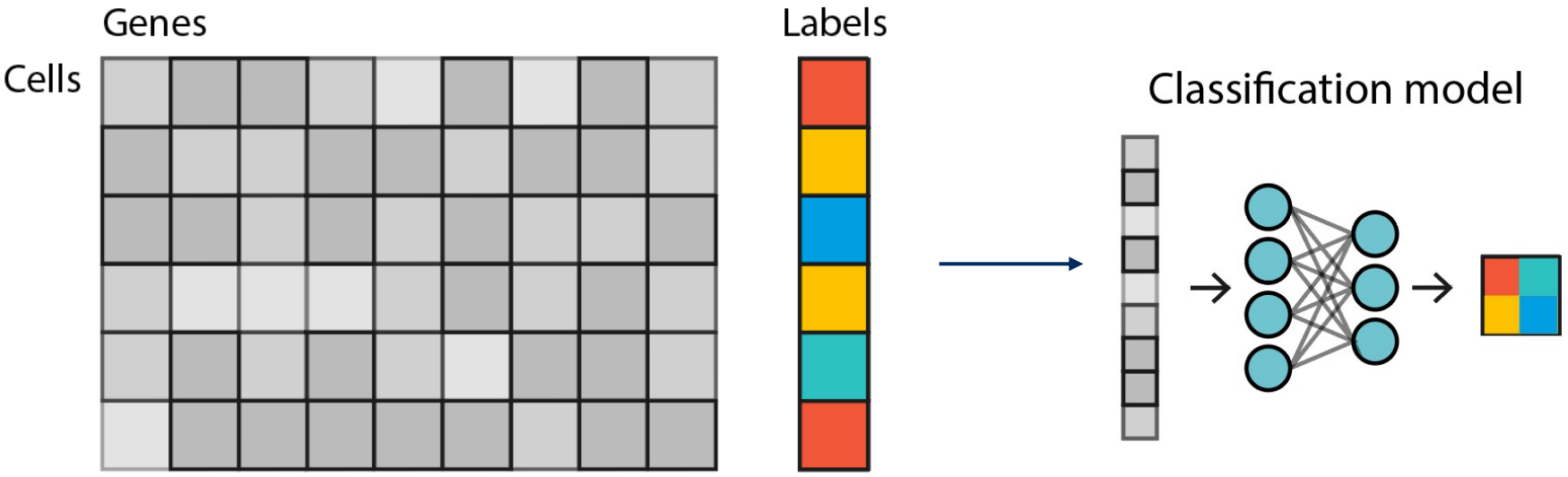
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Uniform Manifold Approximation and Projection (UMAP): a machine learning algorithm used for dimensionality reduction.

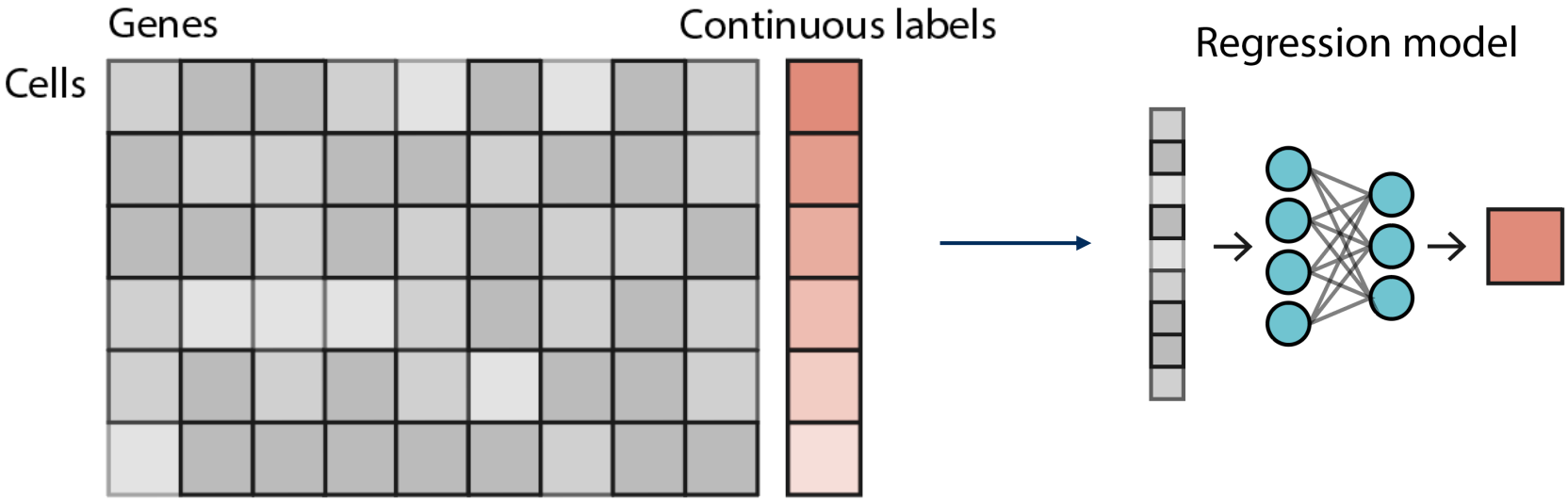
Case Study #1 — Single-Cell RNA-seq

- Goal: Identify cell types
- Data: Expression vectors with cell type labels
- Task: Prediction (supervised)



Case Study #2 — Predicting Drug Response

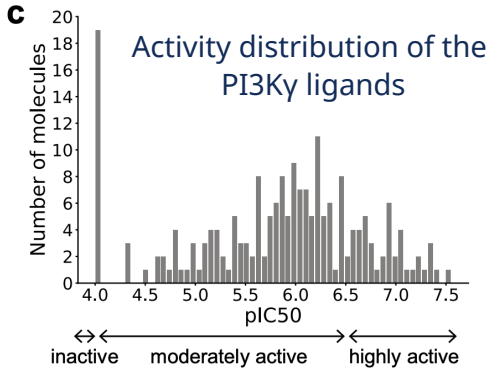
- Goal: Predict cancer cell responses
- Data: Gene expression vectors
- Task: Regression

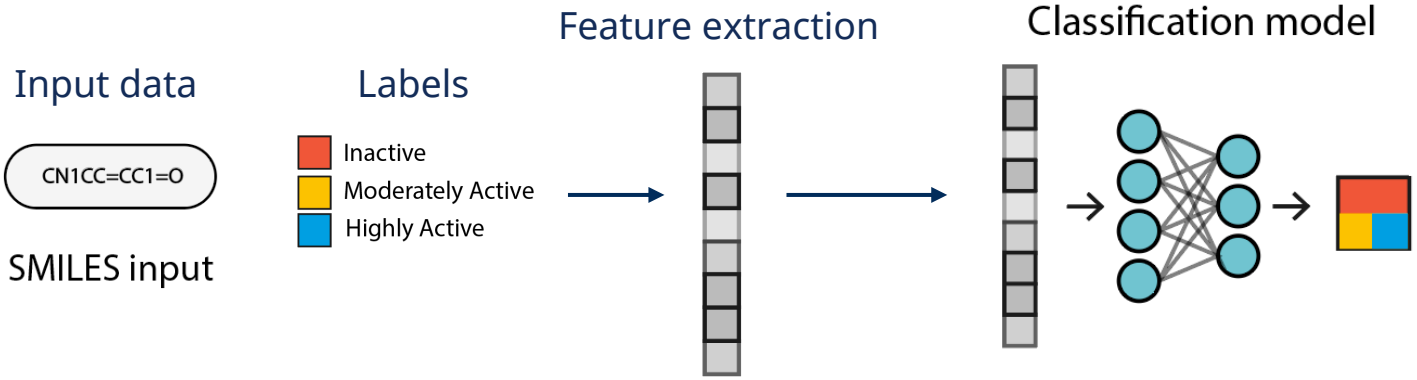


Sinha et al., Nature Cancer (2024)

Case Study #2* — Predicting bioactivity level

- Goal: Prediction of bioactivity against PI3Ky target
- Data: Chemical molecules
- Task: Classification

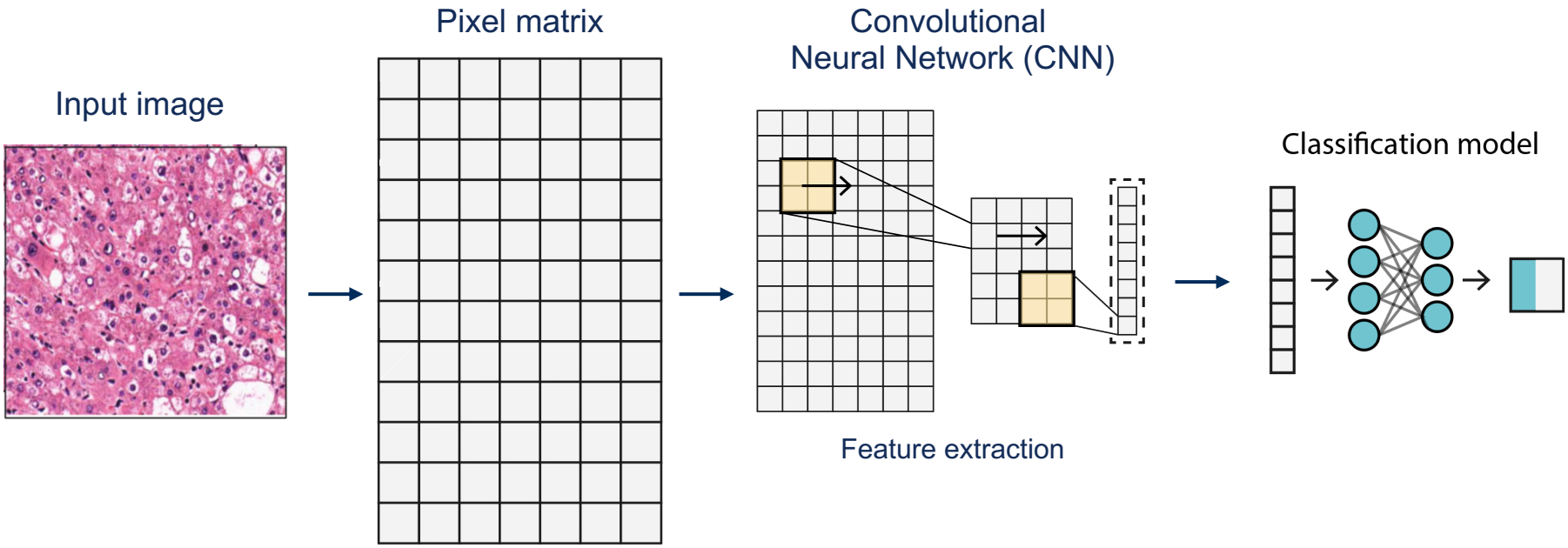




Moret et al., Nature Communications (2021) – input and distribution images are adapted from the paper

Case Study #3 — Image processing

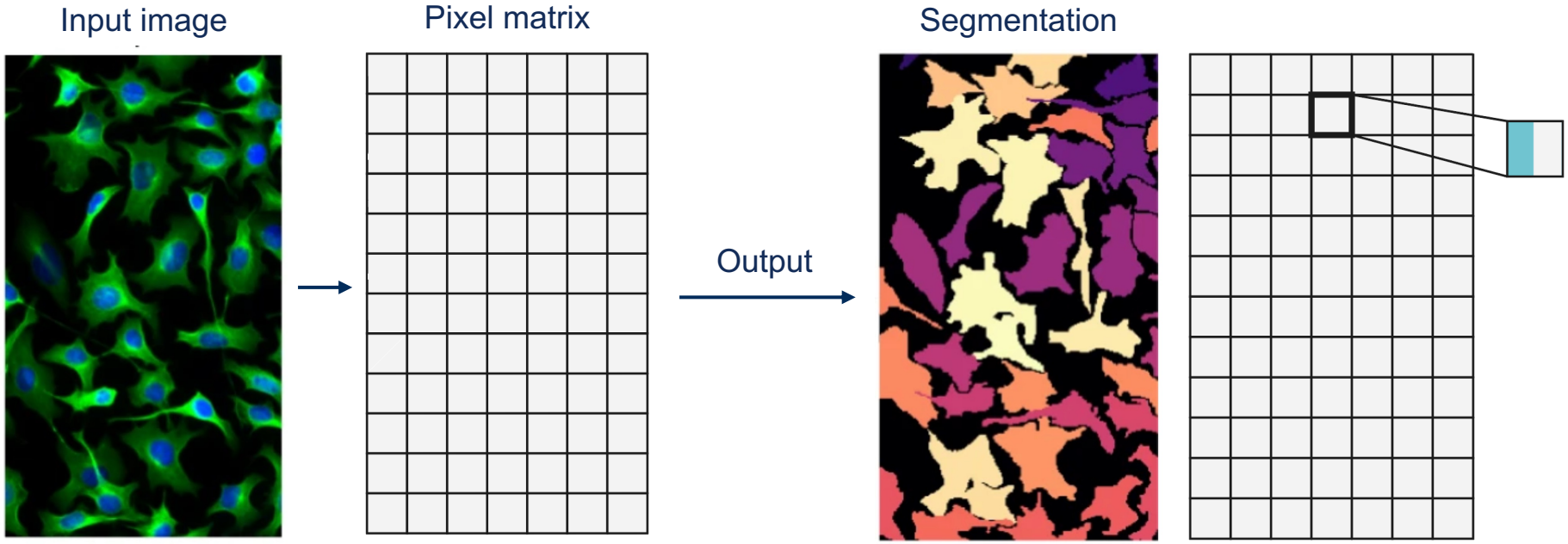
- Goal: Image classification
- Data: Pixel matrices
- Task: Supervised classification



Chen et al., npj Precision Oncology (2024) - input image is adapted from the paper

Case Study #3 — Image processing

- Goal: Identify cellular elements (e.g., cell bodies) in microscopic images
- Data: Pixel matrices
- Task: Supervised classification

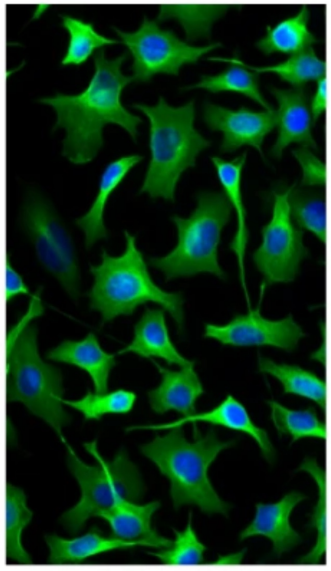


Stringer et al., Nature Methods (2021) - input and segmentation images are adapted from the paper

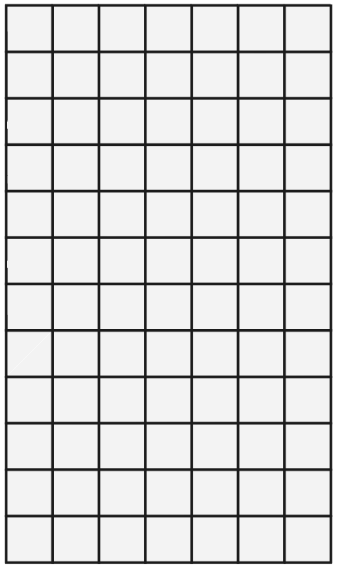
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- Goal: Identify cellular elements (e.g., cell bodies) in microscopic images
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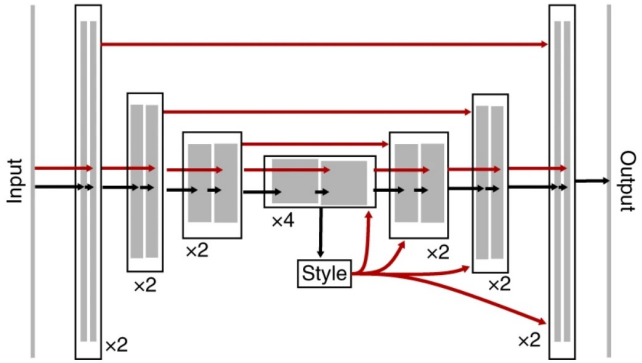
Input image



Pixel matrix



U-Net



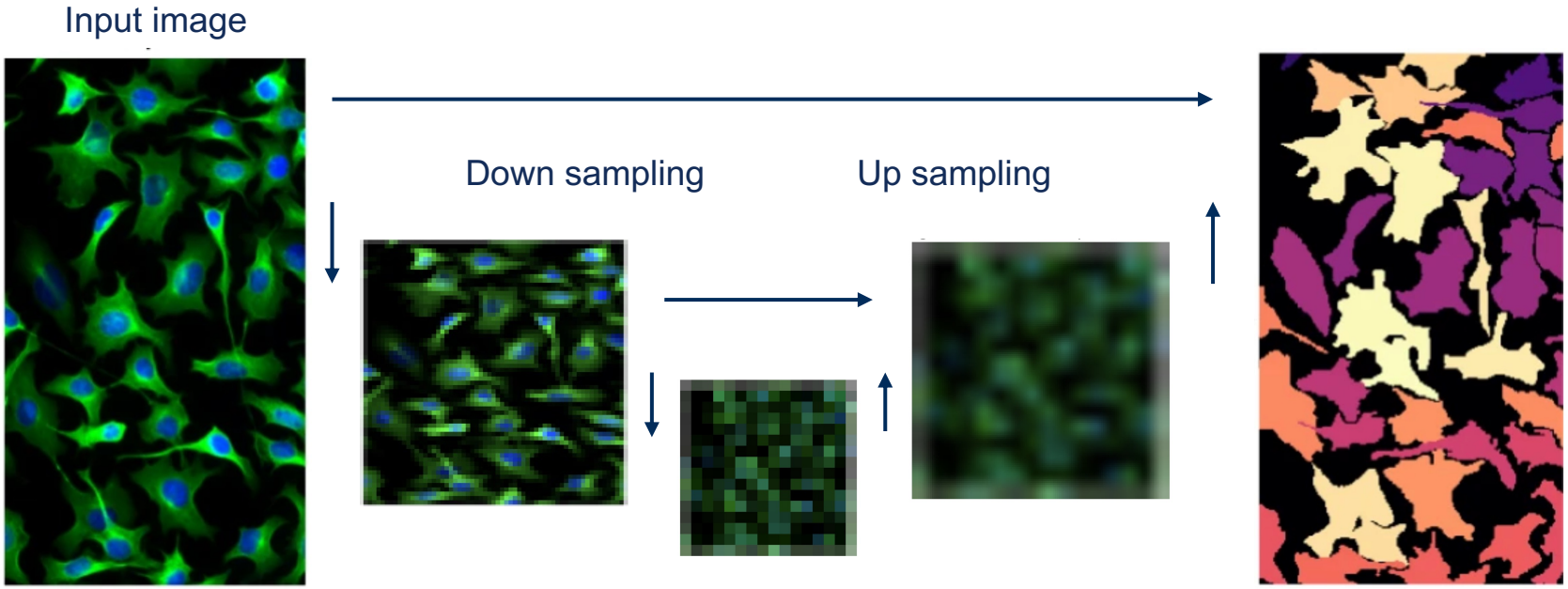
Segmentation



Stringer et al., Nature Methods (2021) - U-Net architecture image is adapted from the paper

Case Study #3 — Image processing

- Goal: Identify cellular elements (e.g., cell bodies) in microscopic images
- Data: Pixel matrices
- Task: Supervised classification

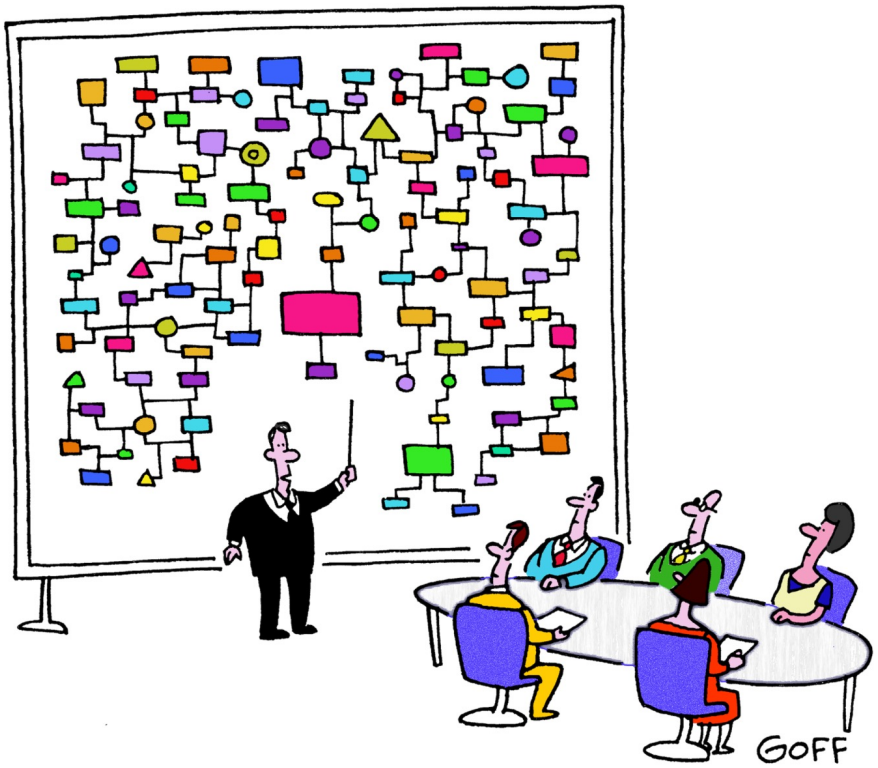


Stringer et al., Nature Methods (2021)

(Down sampled and up sampled images are generated using ChatGPT)

What Makes a Good ML Question?

- ✅ Clear outcome
- ✅ Numeric input
- ✅ Enough data
- ✅ Biological rationale
- 🚫 Tiny datasets
- 🚫 Noisy labels
- 🚫 Confounders



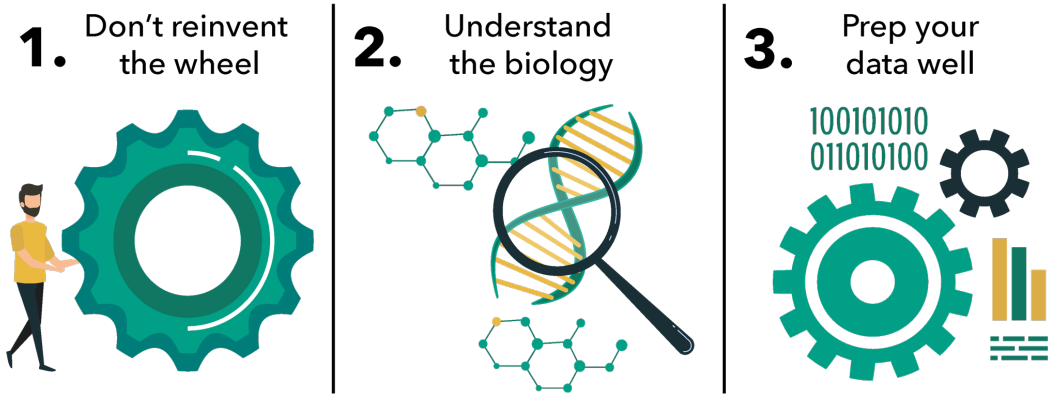
"And that's why we need a computer."

License granting use of this cartoon to B. Villoutreix by T. Goff – USA – March 14, 2014

Credit: https://www.vls3d.com/

Good Practices in Bio-ML

- Don’t reinvent the wheel
- Understand biology first
- Clean/normalize data
- Start with simple models
- Use proper metrics
- Collaborate across fields



Credit: https://lizard.bio/knowledge-hub/best-practices-ml-in-life-sciences

Tools to Explore

- R Ecosystem
  - caret
  - tidymodels
  - randomForest / ranger
  - umap / Rtsne / FactoMineR
- Machine Learning Frameworks (Python)
  - scikit-learn
  - PyTorch
  - TensorFlow
- Single-cell Analysis
  - Seurat (R)
  - Scanpy (Python)
- Image Analysis
  - CellProfiler
  - DeepCell
- Protein Structure & Design
  - AlphaFold
  - RoseTTAFold
  - RFdiffusion

Discussions

- “Think of a biological question & dataset you've seen. Could it be used for ML?”
- What’s the outcome?
- What’s the input?
- What patterns might exist?

Click the following link or scan the code to input your idea:
https://padlet.com/jiaweinohupcc/what-real-world-datasets-have-you-encountered-that-could-be--8tcddl6ilrm5d6zc



Reference

- The content and figures have been credited with links on the relevant slides. These sources are also excellent material for further reading.